Jun Yu (Johnny) Chen

LinkedIn ▪ (310) 962-5942 ▪ junyujohnnychen@gmail.com ▪ Personal page

EDUCATION

UNIVERSITY OF PENNSYLVANIA — **Philadelphia**
Major: *MSE in Data Science, MSE in Computer & Information Science, GPA:3.83* — *Expected May 2027*

UNIVERSITY OF CALIFORNIA, LOS ANGELES — **Los Angeles**
BS in Cognitive Science(Specialization in Computation), Data Science Engineering Minor, GPA:3.94 — *June 2025*

WORK EXPERIENCE & RESEARCH

Tiktok — **Seattle**
Machine Learning Engineer Intern — *May 2026 – Aug 2026*

- Built an **end-to-end** teacher–student **distillation pipeline** for personalized e-commerce intent recognition using query signals and multimodal user behavior, covering streaming offline labeling, data infra, and student models optimized for **150ms** online serving
- Productionized an agentic labeling pipeline on daily-scheduled Spark, SQL, and LLM DAGs with behavioral-signal APIs and **concurrency** controls, generating 40K reasoning-annotated labels per day at **85% accuracy** on human alignment
- Architected a multi-tower model combining personalized gating and DIN sequence attention across 150+ sparse, dense, and sequential features, with class-weighting, parameter-server training, and streaming data-lake ingestion, improved **AUC from 0.52 to 0.83**
- Trained a 0.6B embedding-native Qwen3 decoder from scratch with per-token FFN feature encoders, Perceiver cross-attention, and latent reasoning, followed by policy-gradient RL post-training; production A/B tests showed **GMV and PDP gains of 2.3% and 4.5%**

Wharton Research Data Services (WRDS), Advisor: Dr. Rui Dai — **Philadelphia**
Machine Learning Researcher — *Jan 2026 – Present*

- Built a LangGraph financial research agent turning paper ideas into Python/SQL notebooks for stock-market analysis, replication, and study design; implemented an agent harness with tool orchestration, persistent state, and feedback-driven code generation and validation.
- Developed a retrieval pipeline for ~19.3K finance papers using web scraping, multi-GPU MinerU parsing at **~1.3s/page**, and Amazon S3 ingestion; designed block-aware Qdrant indexing that preserves context across text, equations, and figure captions.

Columbia DAP Lab (Data, Agents, Processes), Advisor: Prof. Zhou Yu — **New York**
Machine Learning Researcher — *April 2025 – Dec 2025*

- Fine-tuned Qwen2.5 and Qwen3 using RLHF framework, combining **SFT finetuning** and **GRPO** (VERL reward) to improve field-level structured generation in Elasticsearch-style job-resume retrieval; achieved **+17% nDCG** over ConFit-V2 baseline
- Built scable SFT training and synthetic data generation pipeline using vLLM, LLaMA-Factory, and DeepSpeed ZeRO-3 on A100s
- Implemented a distillation-based data refinement pipeline (DeepSeek, GLM-4.7) with LLM judge re-ranking to denoise weak labels
- Trained GRPO using VERL with custom reward, curriculum strategy, and loose parsing design, yielding a **+23% recall improvement**.

Job Junior — **Los Angeles**
Co-Founder & Founding Engineer — *Sep 2023 – May 2026*

- Built an end-to-end LLM-based outbound automation system for job and candidate matching; used **Playwright, LangChain, FAISS**, and **QLoRA** to extract job descriptions, match profiles, and generate cold emails with **15%+ response rate** across **1K+ targets**
- Built an asyncio-based ETL pipeline (MongoDB, FastAPI, AWS EC2/Docker) achieving **6x throughput**, enabling daily scraping of 800+ companies with automated delivery via lark and spreadsheet APIs; signed a **$2K+**/month pilot with 5 top recruiting firms.

Generative Machine Learning Research Group, Advisor: Prof. Jiatao Gu, Paper link — **Philadelphia**
Machine Learning Researcher — *March 2024 – Present*

- Engineered a multimodal LLM agent with **RAG memory** and self-reflection loop; supported language/image input to structured blueprint generation and autonomous execution via Node.js control bridge for Minecraft bots; **work published at AAAI W17**.
- Developed a verification pipeline combining multi-view visual feedback and world-state block scanning, enabling ground-truth structural diffing and iterative self-correction with actionable patch generation, outperforming one-shot generation accuracy by 25%

PROJECT EXPERIENCE

Anima Lab, Website — **Philadelphia**
Founder & Founding Engineer — *April 2026 – Present*

- Built and launched a full-stack AI-companion product from 0→1 across Next.js web, native SwiftUI iOS, and React Native Android on a shared TypeScript/Convex backend; reached **1,000+ users**, built cross-platform acquisition channels, and gained **40K+ followers**.
- Architected a **multimodal agent** with image generation, realtime voice call, and self-evolving memory (hybrid retrieval, consolidation, pruning) plus dynamic emotion/attachment state; improved memory-retrieval accuracy by **13%** on LoCoMo and MAU by **20%**.

Synthetic Data Generation with LLM for Improved Depression Prediction, Paper link — **Los Angeles**
Generative Data Science Grad Course — *April 2024 – July 2024*

- Built a **CoT-style** LLaMA3 model with **QLoRA** to generate privacy-preserving patient synopses for PHQ-8 depression prediction, improving downstream **BERT RMSE from 5.01 to 4.39** with validated fidelity via PCA of CLS embeddings; accepted at **CogSci 2024**

TECHNICAL SKILLS

- **Languages & Developer Tools**: Python, C++, SQL, JavaScript/TypeScript, Git, Linux, Bash, REST APIs, Docker, AWS,CI/CD
- **ML & Systems**: PyTorch, Spark, vLLM, FastAPI, LangChain, RAG, TensorFlow, A/B Testing, Distributed Training, Hive, ETL